# CREDIT SCORECARD MODEL ANALYSIS

## *SUMMARY*

This project builds a credit scorecard derived from the well-known Lending Club dataset of US retail loans (CSV download available: https://zenodo.org/records/11295916) using unpenalised and ridge logistic regression in Python.

The data is split between training and testing data, weight-of-evidence and information value scores are used to select variables, and these variables are then tested for multicollinearity using variance inflation factors. The Kolmogorov-Smirnov test statistic is used to determine the optimal threshold for classification and validation is performed using ROC, accuracy, precision, recall, and f1 scores.

The logistic regression output is then converted into scorecard format, which is presented as an interactive Excel workbook. This gives the user the option of centering the scoring on different scores for different odds of default and observing the output given different loan amounts, loan purposes, FICO scores, etc.

## *VARIABLE SELECTION*

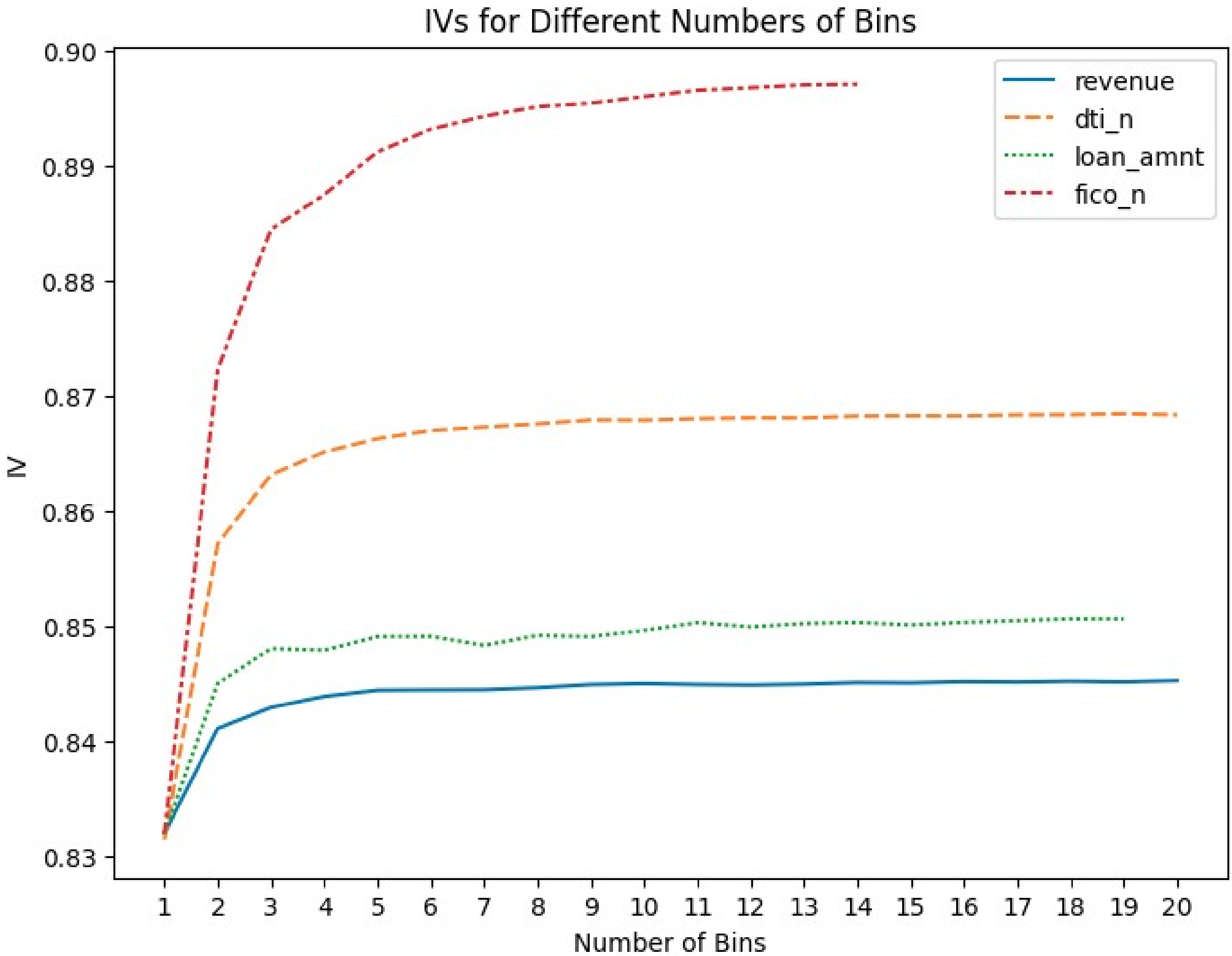
The original variables in the Lending Club dataset are given below:

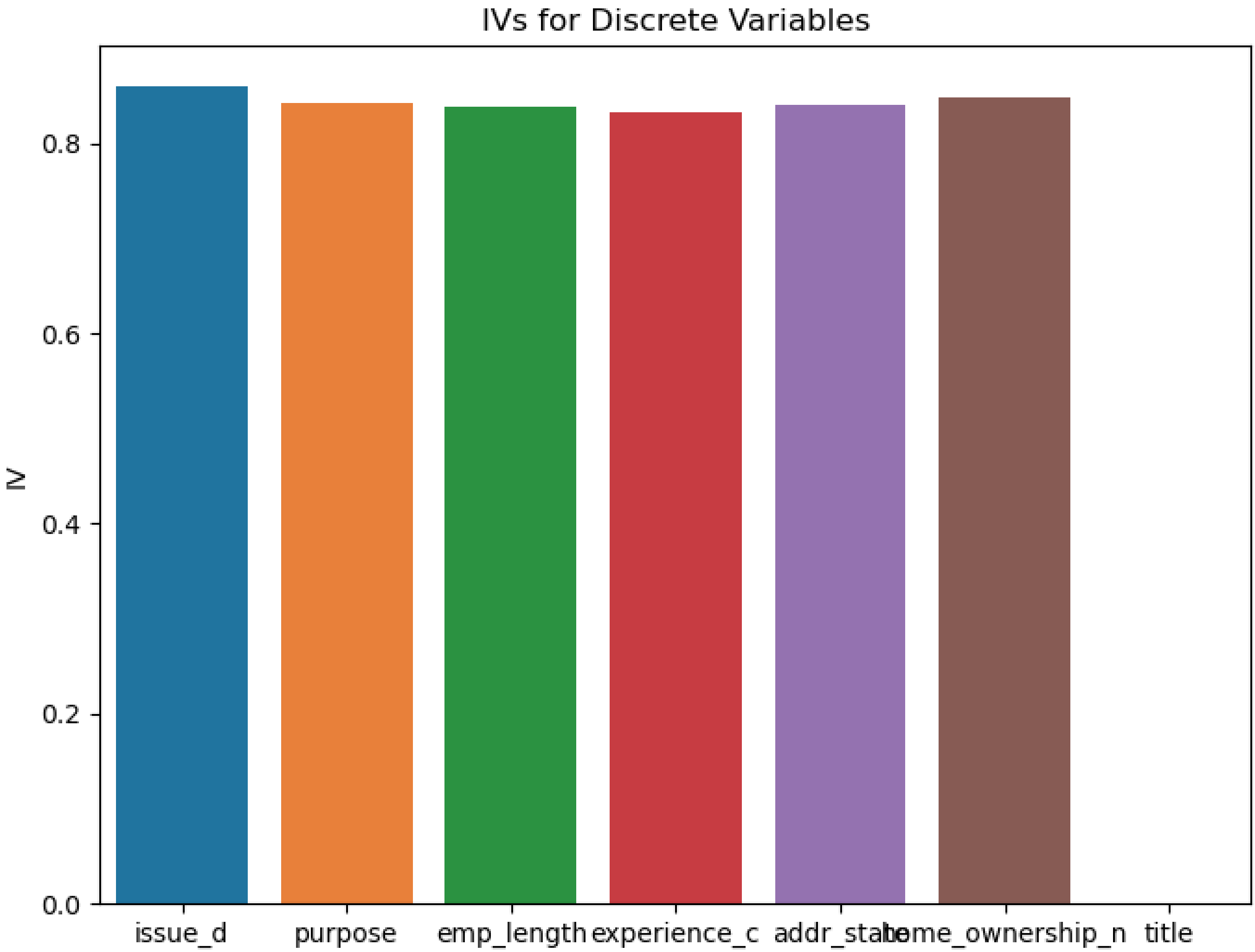
| *Variable Name* | *Data Type* | *Description* |
|---|---|---|
| id | Categorical | ID number for borrower |
| issue_d | Date (month-year) | Date when loan was issued |
| revenue | Numerical | Borrower’s self-declared annual income in USD |
| dti_n | Numerical | Debt to income ratio |
| loan_amnt | Numerical | Amount borrowed in USD |
| fico_n | Numerical | Borrower’s FICO score |
| experience_c | Boolean | Whether the borrower has existing loans with the entity or not (0 = no, 1 = yes) |
| emp_length | Ordinal | Employment length in years (NI = no information) |
| purpose | Categorical | Funding purpose of the loan |
| home_ownership_n | Categorical | Home ownership status: mortgage, rent, own or other |
| addr_state | Categorical | Borrower’s state code |
| zip_code | Categorical | Borrower’s ZIP code (last two digits omitted) |
| Default | Boolean | Whether the borrower defaulted or not (0 = no, 1 = default) |
| title | Categorical | Title of the credit request description provided by the borrower |
| desc | Textual | Description of the credit request |

The following variables were initially dropped for the regression for the following reasons:

| Variable Name | Reason for Dropping |
|---|---|
| id | No relevance for default probability |
| zip_code | Deemed too granular for address information: addr_state sufficient for building a model without overfitting to training data |
| desc | Insufficient data: only 119099/1347681 = 8.8% of entries were non-null. |

From the remaining variables, information values were calculated from weight-of-evidence scores with different numbers of percentile bins tested for the continuous variables. The results are presented in the charts below:





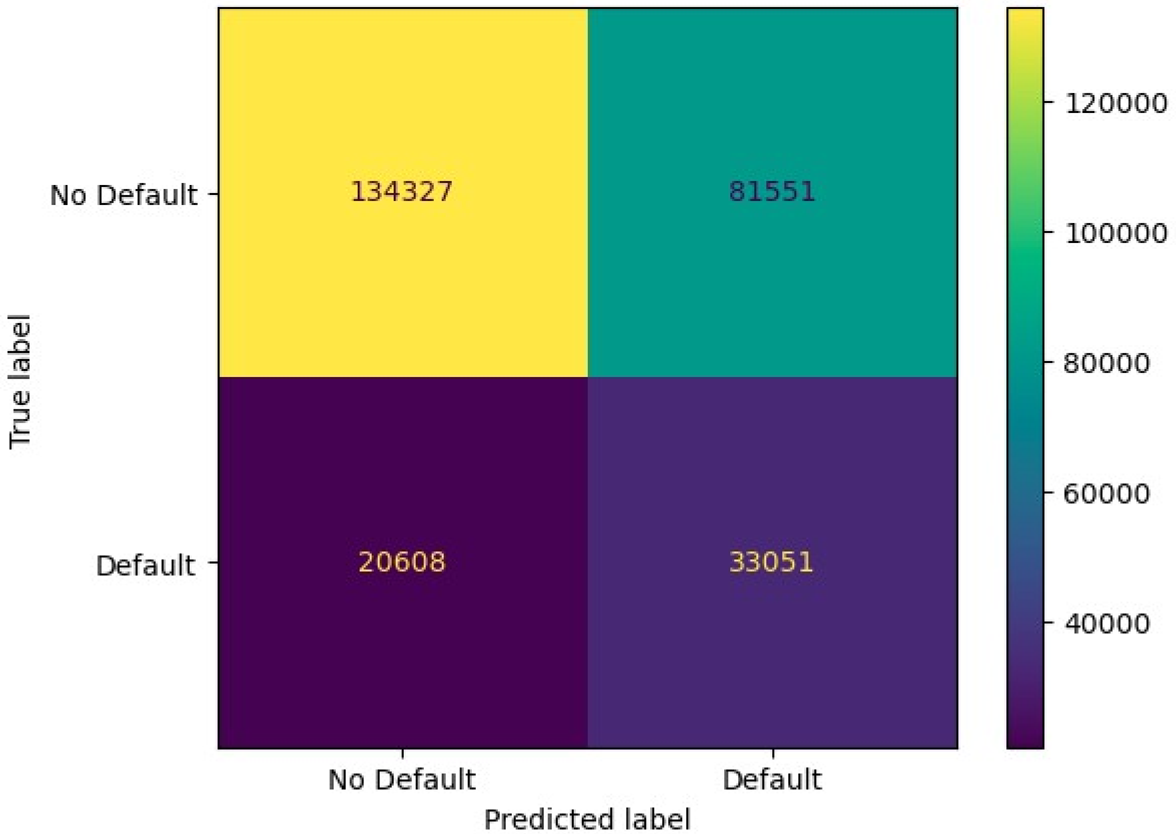

Due to the low IV value for the title variable, title was dropped for the regression stage as well. All other variables were included due to their high IV scores on the training data. 14 bins were chosen for the continuous variables as this was the maximum number of possible equal-percentile-width bins for the fico_n variable (without hitting plus or minus infinity for the weight-of-evidence score) and the other variables level off with low marginal IV gains after 14 bins anyway.

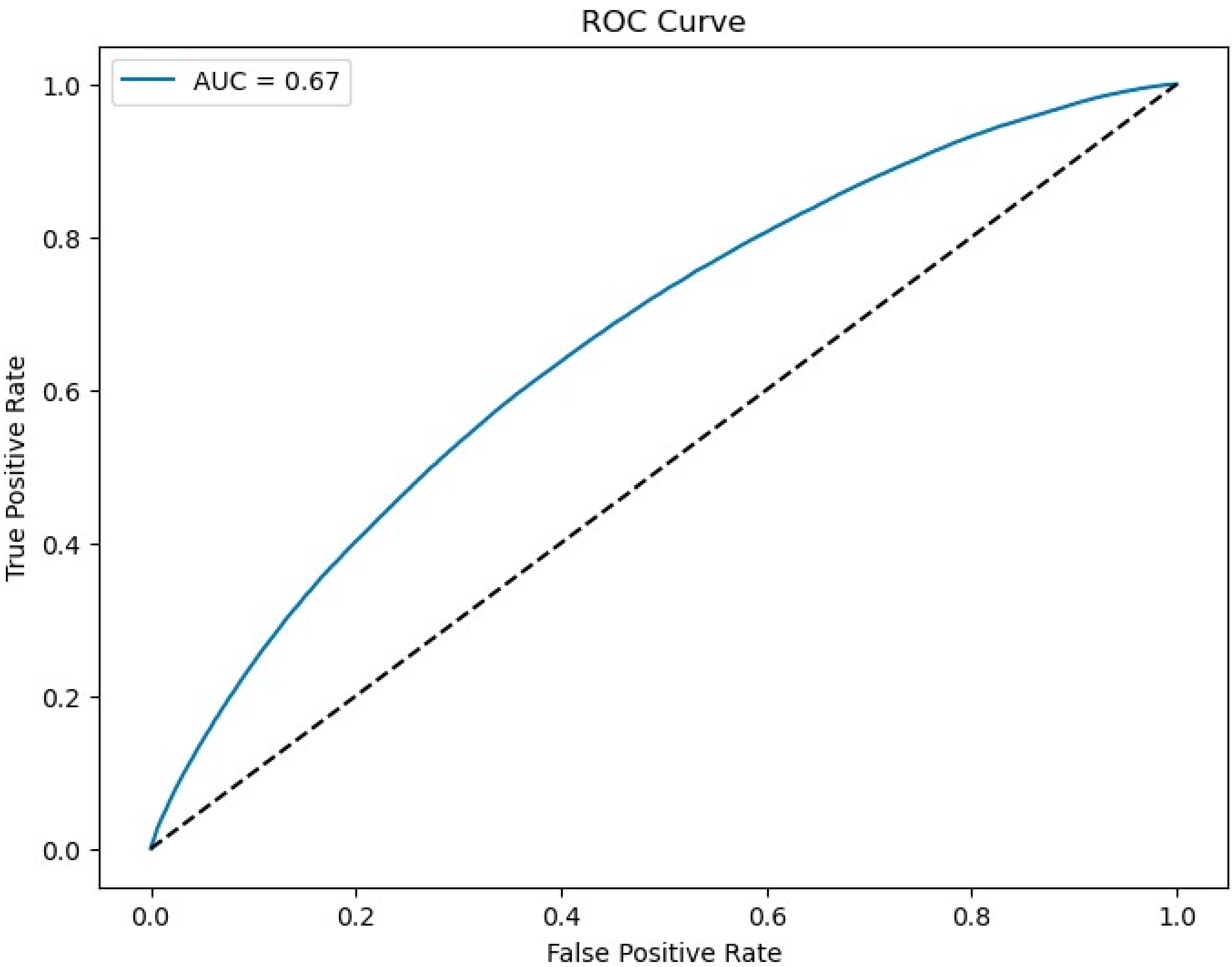
***LOGISTIC REGRESSION***

The data was split between training and testing data in the ratio 80%:20% and the Default=1 class subset was overweighted as the minority class by choosing the "balanced" weighting parameter for logistic regression. Each dataset entry was replaced with bin weight-of-evidence scores and the K-S statistic was used to determine the optimal classification threshold. The model coefficients, ROC, confusion matrix and validation scores from evaluation on the test sample are below:

```
Precision: 0.28839810823545836
Accuracy: 0.6209833900355053
Recall: 0.6159451350192885
F1 Score: 0.39254206679438214
```

| | Variable | Coefficient |
|---|---|---|
| 0 | intercept | 4.332472 |
| 1 | issue_d | -0.919720 |
| 2 | revenue | -1.168640 |
| 3 | dti_n | -0.715134 |
| 4 | loan_amnt | -1.747232 |
| 5 | fico_n | -0.957284 |
| 6 | purpose | -0.865371 |
| 7 | home_ownership_n | -0.827394 |
| 8 | emp_length | -0.916167 |
| 9 | experience_c | 5.998506 |
| 10 | addr_state | -1.008568 |





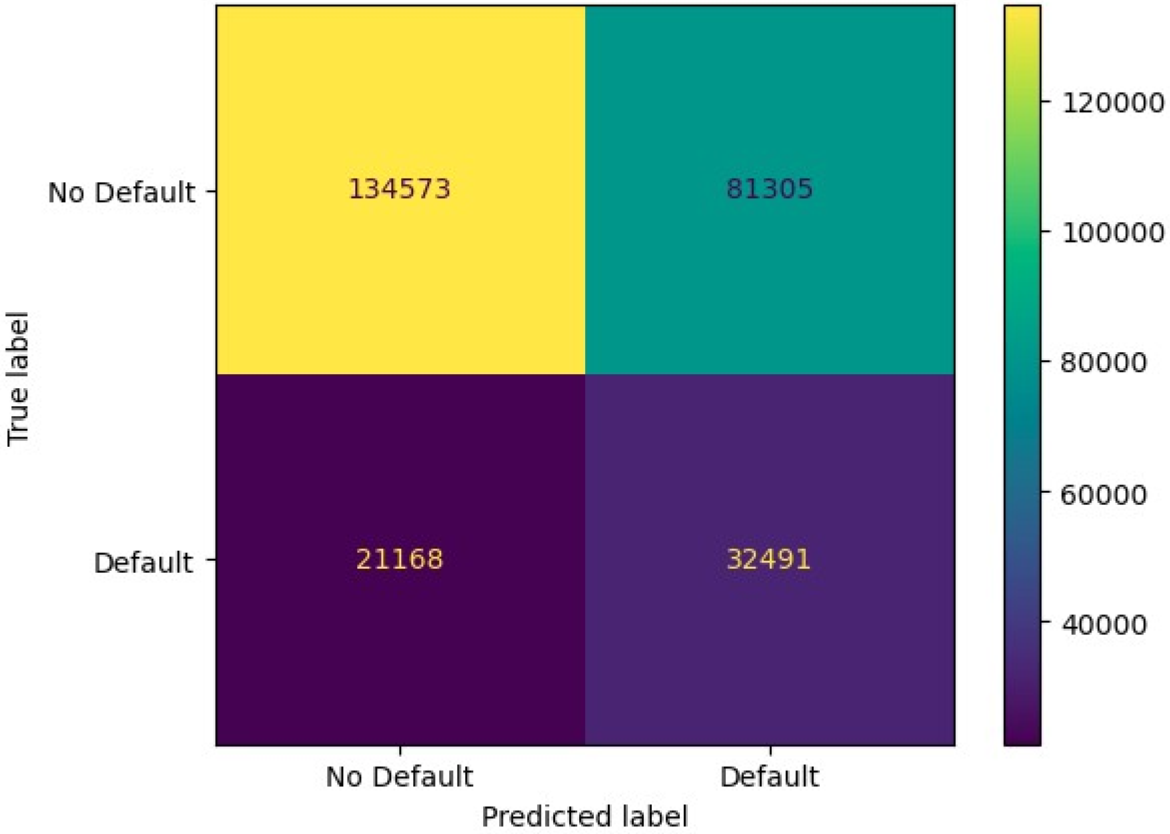

Independent variables were tested for multicollinearity using variance inflation factors, but these returned as normal (implied $R^2$ values for each linear regression as near 0):

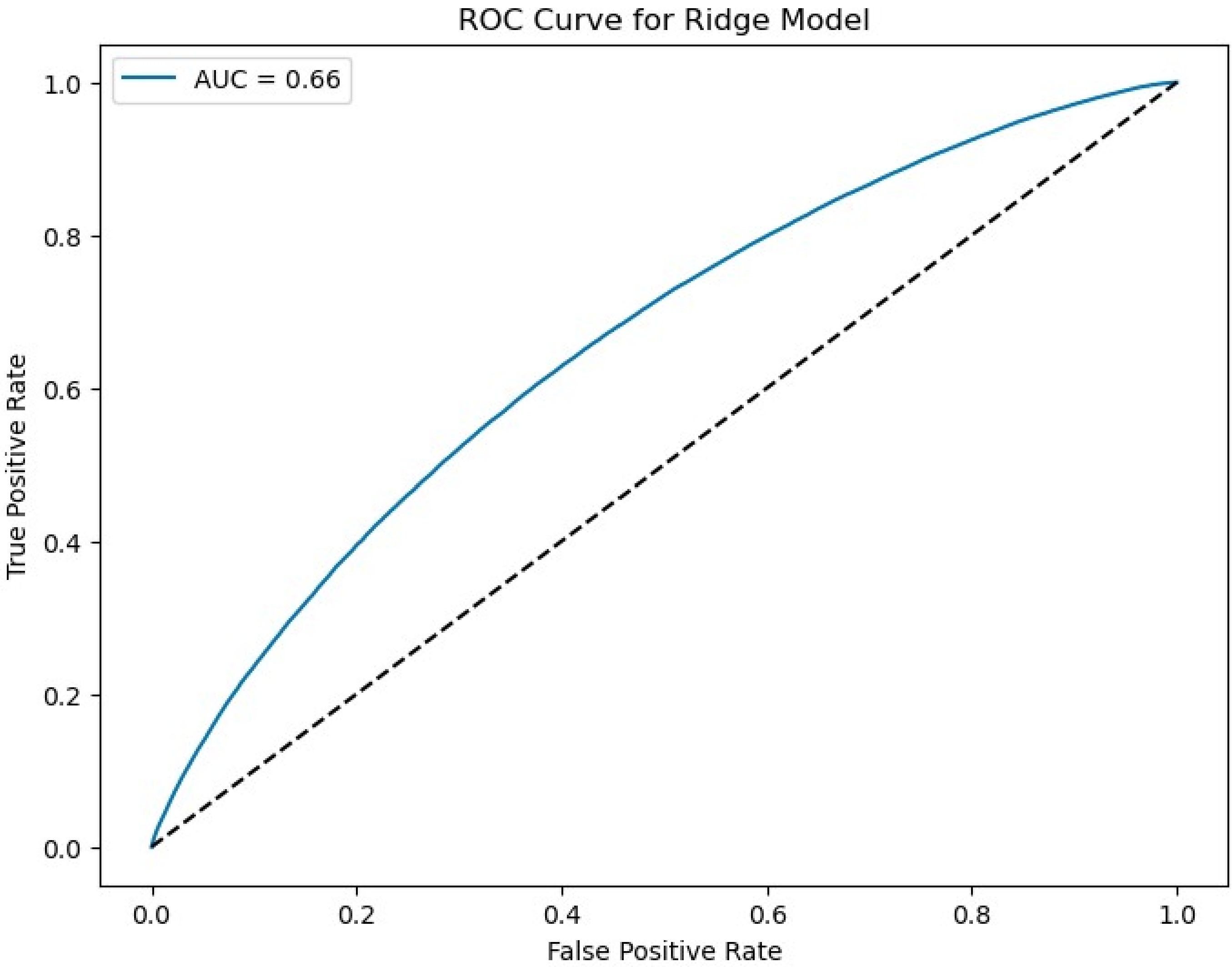
```
revenue             1.385133
loan_amnt           1.246741
home_ownership_n    1.090085
dti_n               1.069949
emp_length          1.038858
fico_n              1.026045
issue_d             1.008591
purpose             1.004741
addr_state          1.000580
experience_c        1.000146
dtype: float64
```

To ensure that the out-of-sample performance of the model could not be improved by reducing the model complexity, a ridge logistic regression model was also trained with a heavy penalisation parameter (C=0.0001). The results of this are below:

```
Precision: 0.28551970192273896
Accuracy: 0.6198184293807529
Recall: 0.6055088615143779
F1 Score: 0.38805649278910753
```

| | Variable | Coefficient |
|---|---|---|
| 0 | intercept | 5.527629 |
| 1 | issue_d | -0.503443 |
| 2 | revenue | -0.346384 |
| 3 | dti_n | -0.566270 |
| 4 | loan_amnt | -0.620617 |
| 5 | fico_n | -0.723210 |
| 6 | purpose | -0.304646 |
| 7 | home_ownership_n | -0.392634 |
| 8 | emp_length | -0.235541 |
| 9 | experience_c | 0.000559 |
| 10 | addr_state | -0.293964 |






Penalising the regression coefficients results in lower validation scores across the board, and (unusually for a ridge model), does not result in fewer effective independent variables. This indicates that all the variables used have predictive power and dramatically reduce model effectiveness if removed. Given the variables included in the regression, the model is optimal.

Both models have accuracy and recall scores better than chance (0.5) but poor precision scores, reflecting a large proportion of false positives. The K-S threshold choice to maximise the gap between the true positive rate and false positive rate is designed to optimise recall, which will inevitably reduce precision. Given that, in this scorecard context, a false positive (predicting a default that does not occur in the end) is less problematic than a false negative (not warning that a borrower is about to default), optimising for recall is preferable.

Incidentally, the poor precision score can be modestly alleviated by not overweighting the defaulting class, but at the expense of a lower recall score. Below are the validation scores for the unpenalised regression without overweighting the defaulted class:

```
Precision: 0.2917993233375965
Accuracy: 0.6306035906016614
Recall: 0.5995266404517415
F1 Score: 0.39254206679438214
```

### ***CREDIT SCORECARD CONVERSION***

An interactive Excel workbook is provided with this project which converts the output of the unpenalised logistic regression into scorecard format. Below is a sample screenshot:

| BASE SCORE | BASE ODDS OF DEFAULT | POINTS TO DOUBLE SCORE | DEFAULT LOG-ODDS | CREDIT SCORE | ISSUE DATE | | ADDRESS STATE | | REVENUE (UPPER BOUNDS) | | TOTAL DEBT / MONTHLY INCOME (UPPER BOUNDS) | | LOAN AMOUNT (UPPER BOUNDS) | | FICO SCORE (UPPER BOUNDS) | | LOAN PURPOSE | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 600 | 50 | 20 | -4.6063 | 620 | Choose: | Aug-07 | Choose: | AZ | Enter: | 50000 | Enter: | 9 | Enter: | 30000 | Enter: | 800 | Choose: | debt_consolidation |
| | | | | | VALUE: | -0.8535 | VALUE: | -1.4161 | VALUE: | -1.4946 | VALUE: | -1.2168 | VALUE: | -2.0808 | VALUE: | -2.2685 | VALUE: | -1.1360 |
| | | | | | Dec-07 | -0.8959 | CT | -1.5709 | 65000 | -1.5761 | 17.64 | -1.0497 | 12000 | -2.4439 | 692 | -1.290 | house | -1.0822 |
| | | | | | Jan-08 | -0.7886 | DC | -1.8936 | 72000 | -1.6295 | 19.31 | -1.0111 | 14900 | -2.1919 | 697 | -1.326 | vacation | -1.2585 |
| | | | | | Feb-08 | -1.2156 | DE | -1.4160 | 80000 | -1.6652 | 21.08 | -0.9705 | 16000 | -2.3245 | 702 | -1.391 | car | -1.5256 |
| | | | | | Mar-08 | -1.1579 | FL | -1.3097 | 88000 | -1.7184 | 23.02 | -0.9178 | 18500 | -2.0965 | 712 | -1.515 | medical | -1.0993 |
| | | | | | Apr-08 | -0.9558 | GA | -1.4990 | 100000 | -1.7752 | 25.2 | -0.8765 | 20400 | -2.1683 | 717 | -1.606 | moving | -1.0417 |
| | | | | | May-08 | -1.1708 | HI | -1.3763 | 115000 | -1.9070 | 27.88 | -0.8063 | 25000 | -2.1362 | 732 | -1.728 | renewable_energy | -1.0360 |
| | | | | | Jun-08 | -1.3833 | IA | -1.5170 | 144000 | -1.8743 | 31.49 | -0.7302 | 30000 | -2.0808 | 752 | -1.927 | wedding | -1.7037 |
| | | | | | Jul-08 | -1.2654 | ID | -1.4968 | 10999200 | -2.0139 | 999 | -0.6114 | 40000 | -1.9638 | 847.5 | -2.268 | educational | -1.1058 |
| | | | | | Aug-08 | -1.8996 | IL | -1.5149 | | | | | | | | | | |
| | | | | | Sep-08 | -1.9408 | IN | -1.3129 | | | | | | | | | | |
| | | | | | Oct-08 | -1.5044 | KS | -1.6006 | | | | | | | | | | |
| | | | | | Nov-08 | -1.3969 | KY | -1.3441 | | | | | | | | | | |
| | | | | | Dec-08 | -1.5360 | LA | -1.2145 | | | | | | | | | | |
| | | | | | Jan-09 | -1.5412 | MA | -1.4715 | | | | | | | | | | |
| | | | | | Feb-09 | -1.5845 | MD | -1.3170 | | | | | | | | | | |
| | | | | | Mar-09 | -1.8065 | ME | -1.9184 | | | | | | | | | | |
| | | | | | Apr-09 | -1.6648 | MI | -1.3887 | | | | | | | | | | |
| | | | | | May-09 | -1.6443 | MN | -1.4171 | | | | | | | | | | |

The user can select a base score with odds of default to center the model, alongside the number of points needed for the score to double (20 is a common value for the latter). Dropdown menus are used to choose settings for the categorical variables, and continuous variables can be entered, so long as the entries are in the range of the training data.

The conditional formatting of each variable provides useful feedback on which settings might lead to greater probability of default:

| *Variable* | *Discussion* |
|---|---|
| Issue Date | • Unsurprisingly, the very newest loans have the lowest default probability ;<br>• However, beyond this, cycles of higher/lower risk are present, possibly correlated to the business cycle. The highest default risk is attributed to the GFC period. |
| Address State | • Maine, Washington DC and Vermont are the lowest risk states;<br>• Mississippi, Nebraska and Arizona are the highest risk states;<br>• This may be attributable to a wealth gap amongst these US states, leading to greater default |

| | |
|---|---|
| | probabilities. |
| Revenue | • Unsurprisingly, there is a strong negative correlation between annual income and probability of default. |
| Total Debt / Monthly Income | • Likewise, there is an unsurprising positive correlation between the debt ratio and default rates. |
| Loan Amount | • Generally, the greater the loan amount, the greater the default probability, but the correlation is less than perfect. |
| FICO Score | • Unsurprisingly, there is a strong negative correlation between FICO score and default risk. |
| Loan Purpose | • Wedding and car loans are the least risky;<br>• Small business loans are by far the most risky. |
| Employment Length | • There is a modest negative correlation between time in employment and default risk, but the greatest risk is attributed to those without employment length information. |
| Home Ownership | • Renters are the most risky;<br>• Mortgage owners are the least risky, surprisingly less risky than those that own their home outright. |
| New Borrower | • Unsurprisingly, those that have multiple borrowings are at greater risk of default. |

## ***REFERENCES***

Ariza-Garzón, M.J., Sanz-Guerrero, M. and Arroyo Gallardo, J. (2024) *Lending Club loan dataset for granting models* (Version 0.1) [Dataset]. Universidad Complutense de Madrid. Available at: https://doi.org/10.5281/zenodo.11295916 (Accessed: 10 July 2026).

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Siddiqi, N. (2006) *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. Hoboken, NJ: John Wiley & Sons.